SOFTWARE REQUIREMENTS SPECIFICATION (SRS) & SYSTEM ARCHITECTURE BLUEPRINT

BD-LawAssist: AI Legal Research, Document Analysis & Voice Dictation Platform (MVP)

Decoupled API-First Backend, Temporal Statutory Engine (bdlaws.minlaw.gov.bd), Bilingual OCR & Legal Speech-to-Text

Document Scope	Full Minimum Viable Product (MVP) Specification	**Target Jurisdiction**	Laws of Bangladesh (Ministry of Law Corpus)
Architecture Pattern	Decoupled API Backend (Web + Mobile Clients)	**Primary Personas**	General Citizen, Lawyer (Advocate), Judge
Primary Data Source	bdlaws.minlaw.gov.bd (Automated Sync & Diffing)	**Core AI Pipelines**	Citation-Backed RAG, Bilingual OCR, Legal STT

1. Executive Summary & System Overview

BD-LawAssist is a specialized legal intelligence and case-file assistance platform built upon the authoritative statutory corpus of the Ministry of Law, Justice and Parliamentary Affairs (`https://bdlaws.minlaw.gov.bd/laws-of-bangladesh.html`). Because laws in Bangladesh undergo frequent legislative amendments via Acts and Ordinances, standard static AI models suffer from outdated knowledge and hallucinations. BD-LawAssist solves this through a deterministic **Temporal Statutory Version Control Engine** paired with a strict **Citation-Backed Retrieval-Augmented Generation (RAG)** pipeline.

The platform is engineered around a **Decoupled API-First Backend** that serves both a responsive Web Application (optimized for courtroom preparation, split-screen statutory comparison, and long-form drafting) and a cross-platform Mobile Application (optimized for camera-based FIR/document scanning and push-to-talk voice queries). It serves three distinct user tiers—**General Citizens**, **Lawyers**, and **Judges**—and includes a bilingual (Bangla/English) **Document & FIR OCR Analyzer** as well as a real-time **Voice-to-Text (STT) Dictation Engine** for querying laws and drafting case files hands-free.

2. Decoupled System Architecture (Web + Mobile + Core Backend)

To support Web and Mobile applications from a single source of truth while isolating heavy compute workloads (OCR, Speech-to-Text audio streaming, and vector search), the system follows an asynchronous, API-first modular monolith architecture.

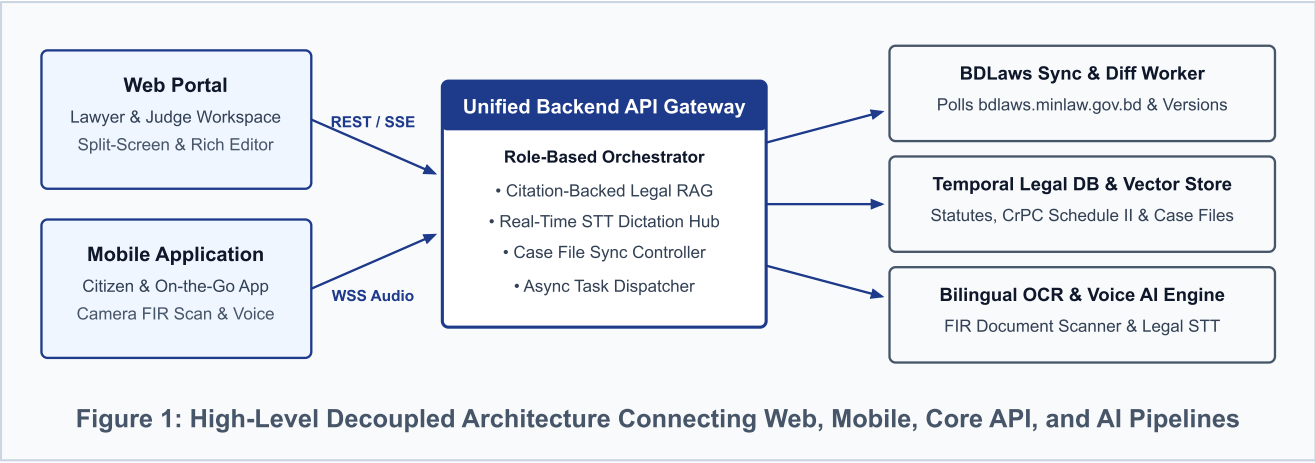


Figure 1: High-Level Decoupled Architecture Connecting Web, Mobile, Core API, and AI Pipelines

Capability	General User (Citizen)	Lawyer (Advocate)	Judge (Judicial Officer)
Response Tone & Style	Plain, empathetic Bangla or English; explains rights and immediate procedural steps without jargon.	Adversarial, strategic, analytical; highlights defense loopholes, prosecution burdens, and cross-references.	Strictly neutral, bench-memo style; focuses on statutory compliance, jurisdiction, and admissibility.
Statutory Querying	Natural language Q&A (e.g., "What to do if police refuse to take an FIR?" or "Punishment for check dishonor").	Exact section lookup, historical law version comparison, limitation period calculation, linked Evidence Act rules.	Rapid verification of sentencing minimums/maximums, fine ceilings, and whether an offense is compoundable.
FIR / Document Upload Analysis	Decodes listed sections into plain table: Bailable vs. Non-Bailable, Cognizable vs. Non-Cognizable, and max penalty.	Audits FIR narrative against statutory ingredients; flags delay in lodging FIR, missing elements, and bail/quashment grounds.	Generates an objective Charge-Framing Matrix mapping prosecution allegations to statutory elements and CrPC compliance.
Voice-to-Text & Case Drafting	Voice search for legal questions; voice-assisted filling of basic General Diary (GD) or complaint applications.	Hands-free dictation of bail petitions, legal notices, cross-examination points, and client interview notes into Case Files.	Hands-free dictation of bench notes, order sheets, issue-framing checklists, and judgment summaries.

4. Module A: BDLaws Scraper & Temporal Version Control Engine

The portal at `https://bdlaws.minlaw.gov.bd/laws-of-bangladesh.html` publishes all Acts of Parliament, Ordinances, and President's Orders in chronological and alphabetical indexes. When an existing law is amended by a new Act, the portal updates the consolidated text and inserts legislative footnotes detailing the substitution, insertion, or repeal.

4.1 Functional Requirements for Automated Corpus Sync

- **FR-A1 (Full-Act Print View Scraping):** Instead of issuing thousands of HTTP requests for individual section URLs, the scraper worker shall target the consolidated print endpoint (`https://bdlaws.minlaw.gov.bd/act-print-{act_id}.html`) to retrieve the complete DOM of an Act—including Preamble, Chapters, Section Numbers, Marginal Headings, Body Text, and Amendment Footnotes—in a single request.
- **FR-A2 (Section-Level SHA-256 Diffing):** During scheduled weekly crawls (or manual admin triggers upon gazette publication), the parser shall normalize Unicode/whitespace for each section and compute a `SHA-256` hash. Only sections whose hash differs from the currently active database record shall trigger re-versioning and re-embedding.
- **FR-A3 (Slowly Changing Dimension Type-2 Versioning):** When an amended section is detected, the system shall never overwrite the old law. It shall mark the existing row as `is_current = FALSE` with a `valid_to` date, and insert a new row with `version_number = N + 1`, `is_current = TRUE`, `valid_from` parsed from the amendment footnote, and a fresh vector embedding.
- **FR-A4 (Point-in-Time Legal Retrieval):** Because criminal liability under Article 35(1) of the Constitution of Bangladesh is governed by the law in force at the time of the act charged as an offense, Lawyers and Judges can specify an `offense_date` in their Case File. The query engine will then retrieve the version of the statute that was active on that exact historical date.

4.2 Automated Statutory Sync & Temporal Version Control Diagram

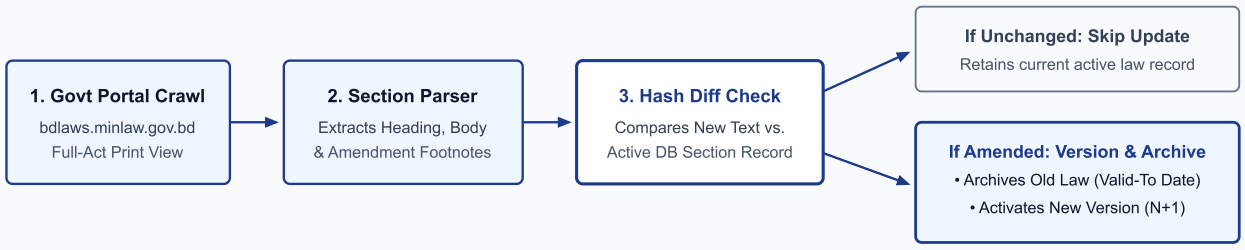


Figure 2: Automated Statutory Diffing and SCD Type-2 Temporal Versioning Pipeline

5. Module B: Document & FIR Scanning Pipeline (OCR + Legal NER)

Users across all three personas can upload PDFs, DOCX files, or smartphone camera images of legal documents—most notably **BP Form No. 24 (First Information Report / এজাহার)**, Charge Sheets (অভিযোগপত্র), Seizure Lists (জব্দ তালিকা), Bail Petitions, and Civil Plaints.

5.1 Functional Requirements for Document Processing

- **FR-B1 (Mobile Image Pre-Processing):** The mobile app scanner shall apply perspective correction, contrast enhancement, and binarization before uploading multipart images to the backend API.
- **FR-B2 (Bilingual OCR & Legacy Font Normalization):** The backend OCR service shall extract both Bengali and English text from printed and legible handwritten documents. If legacy ANSI/Bijoy encoding artifacts are detected in uploaded PDFs/DOCX files, a deterministic converter shall normalize them into standard UTF-8 Bangla Unicode prior to entity extraction.
- **FR-B3 (Deterministic Section Extraction & CrPC Schedule Join):** The pipeline shall use a hybrid Regex + LLM Named Entity Recognition (NER) pass to extract cited acts and section numbers (handling both Bangla numerals like `ধারা ৩০২/৩৪` and English numerals like `Section 302/34 Penal Code`). Extracted sections are immediately joined against the `crpc_schedule_metadata` database table to guarantee 100% factual accuracy regarding whether the offense is **Bailable**, **Cognizable**, or **Compoundable**.
- **FR-B4 (Discrepancy & Ingredient Analysis):** For Lawyers and Judges, the LLM compares the extracted narrative ("Fact of the Case") against the verbatim statutory requirements of the cited sections, highlighting missing legal elements or unexplained delays between the `time_of_occurrence` and `time_of_reporting`.

5.2 FIR & Legal Document Extraction Workflow & Output Structure

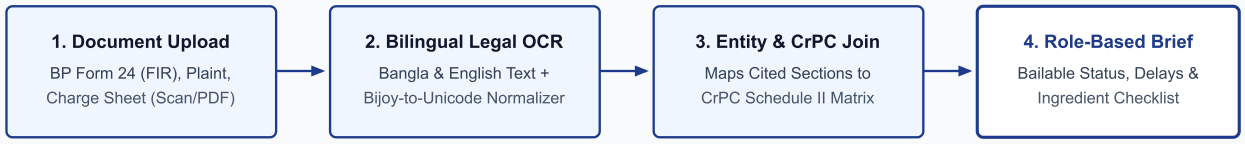


Figure 3: End-to-End Document & FIR Processing Pipeline

Extracted Category	Captured Data Fields	Automated Legal Verification Output
Case & Thana Metadata	Police Station (Thana), District, FIR/G.R. Number, Date & Time of Occurrence vs. Reporting.	Calculates exact reporting delay in hours and flags unexplained delays for defense scrutiny.
Cited Sections Breakdown	Act Name and Section Numbers (e.g., Penal Code Sections 406, 420, 302/34).	Deterministic lookup from CrPC Schedule II: **Bailable vs. Non-Bailable**, **Cognizable**, **Compoundable**, Max Punishment, and Trial Court.
Role-Specific Assessment	Complainant narrative, accused roles, witness list, and seizure references.	**Citizen:** Plain-language rights guide. **Lawyer:** Missing statutory elements & bail grounds. **Judge:** Charge-framing ingredient matrix.

6. Module C: Real-Time Voice-to-Text (STT) & Case File Dictation Engine

Legal practitioners and citizens in Bangladesh frequently communicate in **code-switched Bangla and English** (e.g., speaking a sentence in Bengali while referencing English statutory terms like *"Section 561A CrPC অনুযায়ী quashment petition"*, *"prima facie case"*, *"ad-interim injunction"*, or Revenue/Land terms like *"খতিয়ান, দাগ নম্বর, নামজারি"*). Off-the-shelf browser voice typing fails on this vocabulary.

6.1 Voice Architecture & Functional Requirements

- **FR-C1 (Mode 1 — Push-to-Talk Voice Query):** On both Mobile and Web, users can record a voice question (`m4a`/`webm` Opus audio). The backend transcribes the audio, normalizes legal section numbers, and pipes the query directly into the RAG engine, returning both text and optional Text-to-Speech (TTS) audio summary for General Users.
- **FR-C2 (Mode 2 — Live WebSocket Case File Dictation):** Inside the Case File Editor, Lawyers and Judges can toggle continuous microphone dictation. The client streams 250ms PCM/Opus audio frames over `WSS /api/v1/voice/stream-dictation`.
- **FR-C3 (Context-Biased ASR + Legal Glossary Post-Processor):** Audio chunks are transcribed via Whisper Large-v3 initialized with a custom bilingual prompt glossary (`initial_prompt`) containing Bangladeshi legal terms. Before committing a sentence to the document editor, a fast LLM correction pass fixes phonetic homophones (e.g., correcting *"ফৌজদারি কার্যবিধি"*, *"নালিশী ভূমি"*, *"এজাহার"*, *"চার্জশিট"*) and formats spoken punctuation commands.
- **FR-C4 (Voice-Triggered AI Drafting Commands):** While dictating into a Case File, users can utter voice commands such as *"Insert Section 498 CrPC bail header"* or *"Summarize the uploaded FIR facts here"*, and the editor dynamically inserts the structured legal block at the cursor position.

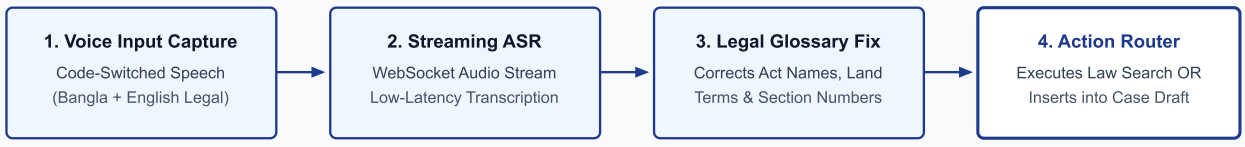


Figure 4: Bilingual Voice-to-Text (STT) Query and Live Case File Dictation Pipeline

7. Data Architecture & Entity Relationship Design

The platform uses a unified relational and vector database architecture to manage user roles, temporal statutory versions, deterministic CrPC Schedule II rules, and multi-device Case Files.

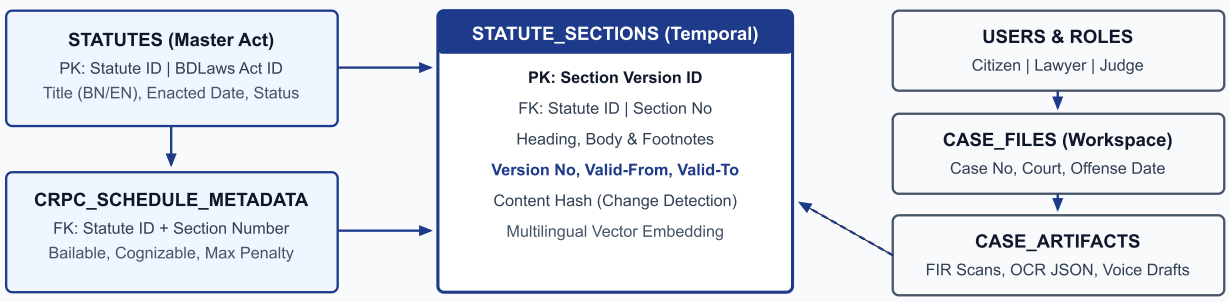


Figure 5: Core Relational and Temporal Vector Database Entity Relationships

Database Entity	Key Attributes	Role in MVP System
Users	User ID, Full Name, Role (Citizen/Lawyer/Judge), Bar/Judicial ID, Language Preference.	Controls workspace permissions, tenant data isolation, and persona-specific AI framing.
Statutes	Statute ID, BDLaws Act ID, Act Number, Title (Bangla & English), Enacted Date, Repealed Flag.	Master catalog of all Bangladesh laws synchronized from the official Ministry of Law portal.
Statute Sections	Section ID, Statute ID, Section Number, Heading, Body Text, Footnotes, Version Number, Valid From/To, Vector Embedding.	Stores current and historical versions of every law section for semantic search and point-in-time retrieval.
CrPC Schedule Metadata	Statute ID, Section Number, Cognizable Flag, Bailable Flag, Compoundable Flag, Max Punishment, Trial Court.	Provides zero-hallucination deterministic rules when analyzing criminal sections extracted from FIRs.
Case Files & Artifacts	Case ID, Owner User ID, Case Number, Offense Date, Uploaded File URL, Extracted OCR Entities, Rich-Text Drafts.	Synchronizes scanned FIRs, voice recordings, and drafted petitions across Web and Mobile devices.

8. REST, SSE & WebSocket API Contract

The backend exposes a strictly versioned API contract so the Web (Next.js) and Mobile (Flutter/React Native) applications share 100% of business logic.

Method	Endpoint URI	Request / Response Payload & Behavior
`POST`	`/api/v1/auth/token`	Authenticates credentials; returns access/refresh JWT with embedded `role` (`CITIZEN` | `LAWYER` | `JUDGE`).

Method	Endpoint URI	Request / Response Payload & Behavior
`POST (SSE)`	`/api/v1/assistant/query`	Accepts `{ query_text, case_file_id?, target_date?, language }`. Performs hybrid vector + keyword retrieval and streams citation-backed tokens via Server-Sent Events.
`GET`	`/api/v1/statutes/{act_id}/sections/{sec_no}`	Returns verbatim text from `bdlaws`, active status, amendment footnotes, and version history diff for the side-panel verifier.
`POST`	`/api/v1/cases/{id}/documents/upload`	Multipart upload of FIR/Charge Sheet image or PDF. Dispatches Celery OCR + Legal NER job; returns `{ job_id, artifact_id }`.
`GET`	`/api/v1/cases/{id}/documents/{doc_id}`	Returns OCR text, extracted metadata, deterministic CrPC Schedule II table (Bailable/Cognizable), and persona-specific legal analysis.
`POST`	`/api/v1/voice/query-transcribe`	Uploads short push-to-talk audio clip; returns cleaned Bangla/English legal transcript and triggers immediate statutory search.
`WSS`	`/api/v1/voice/stream-dictation`	Bi-directional WebSocket accepting binary audio frames from Web/Mobile mic and emitting real-time legal-corrected text deltas into the Case File editor.
`POST`	`/api/v1/cases/{id}/artifacts/{art_id}/export`	Compiles a voice-dictated or AI-assisted draft into a court-formatted DOCX or PDF document with standard Bangladeshi legal margins.

9. Non-Functional Requirements, Security & Guardrails

> **Mandatory Zero-Hallucination & Citation Rule:** The LLM orchestration layer is strictly prohibited from answering statutory questions from pre-trained weights alone. Every legal statement must reference a retrieved `statute_sections.id` record and render a clickable inline tag (e.g., `[Section 497, CrPC 1898]`) that opens the verbatim government text. If no matching section exists in the local `bdlaws` corpus, the system must explicitly respond that no verified statutory provision was found.

- **NFR-1 (Cross-Lingual Semantic Retrieval):** Because laws enacted prior to 1987 in Bangladesh are predominantly in English (e.g., Penal Code 1860, Evidence Act 1872, Contract Act 1872) while newer laws (e.g., Nari-O-Shishu Nirjatan Daman Ain 2000, Cyber Security Act 2023) are in Bengali, the vector pipeline must use a cross-lingual embedding model (`BAAI/bge-m3`) paired with bilingual query expansion so a Bangla query accurately retrieves English statutes and explains them in Bangla.
- **NFR-2 (Tenant Isolation & Legal Privilege):** All uploaded FIRs, case files, and voice recordings are strictly isolated per `owner_user_id` using Row-Level Security (RLS) in PostgreSQL and presigned S3 URLs with 15-minute expiry. User case data is never indexed into the global RAG knowledge base.
- **NFR-3 (Performance & Latency Targets):**
  - Statutory RAG first-token latency (SSE): `≤ 1.5 seconds`.
  - Real-time WebSocket Voice Dictation transcript delta latency: `≤ 800 milliseconds`.
  - Single-page FIR camera scan OCR + Schedule II breakdown: `≤ 6.0 seconds`.

10. 12-Week MVP Implementation & Delivery Roadmap

Phase & Timeline	Milestone Name	Key Engineering Deliverables
Phase 1 Weeks 1–3	Core Backend & BDLaws Temporal Engine	Initialize FastAPI + PostgreSQL/ `pgvector`; build the `act-print-{id}.html` scraper and SHA-256 diffing worker; ingest the Top 50 foundational Bangladesh statutes (Criminal, Civil, Family, Land, Cyber, Labour) + populate the deterministic CrPC Schedule II table.
Phase 2 Weeks 4–6	Persona-Aware RAG & Web Portal	Implement hybrid search (HNSW vector + trigram section matcher); build role-specific system prompts for Citizen, Lawyer, and Judge; launch Next.js Web workspace with split-screen verbatim statute verification.
Phase 3 Weeks 7–9	FIR/Document OCR & Voice-to-Text Pipeline	Integrate bilingual OCR and Bijoy-to-Unicode normalizer; build FIR entity extractor and automatic Bailable/Cognizable lookup; deploy WebSocket voice dictation endpoint with legal glossary post-correction.
Phase 4 Weeks 10–12	Mobile App, Full Corpus Sync & Pilot Testing	Connect Flutter/React Native mobile app (camera document scanner + push-to-talk voice assistant) to the unified FastAPI backend; expand scraper to all ~1,480+ laws on `bdlaws.minlaw.gov.bd`; conduct UAT with practicing advocates.